Student Name: _____

Class Name: **ALEKS Integrated Mathematics II**

Number of Questions: **34**

Instructor Name: **Bodanske, Daniel**

Question 1 of 34

In a scale drawing of a house, 1 inch represents 9 feet.

Scale
1 in : 9 ft

Answer the following.

(a)	The length of the real house is 27 feet. What is the length of the house in the scale drawing? _____ inches
(b)	In the scale drawing, the height of the house is 8 inches. What is the height of the real house? _____ feet

Question 2 of 34

The table below gives the dimensions of a painting and a scale drawing of the painting.

Find the scale factor of the drawing. (The scale factor to get the drawing from the painting.)
Write your answer as a fraction in simplest form.

	Painting	**Drawing**
Length (inches)	45	9
Width (inches)	25	5

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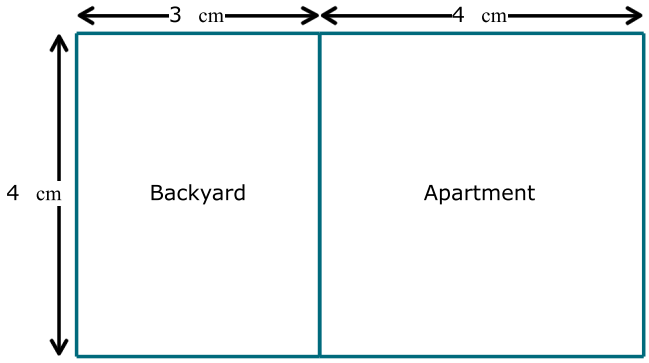
Question 3 of 34

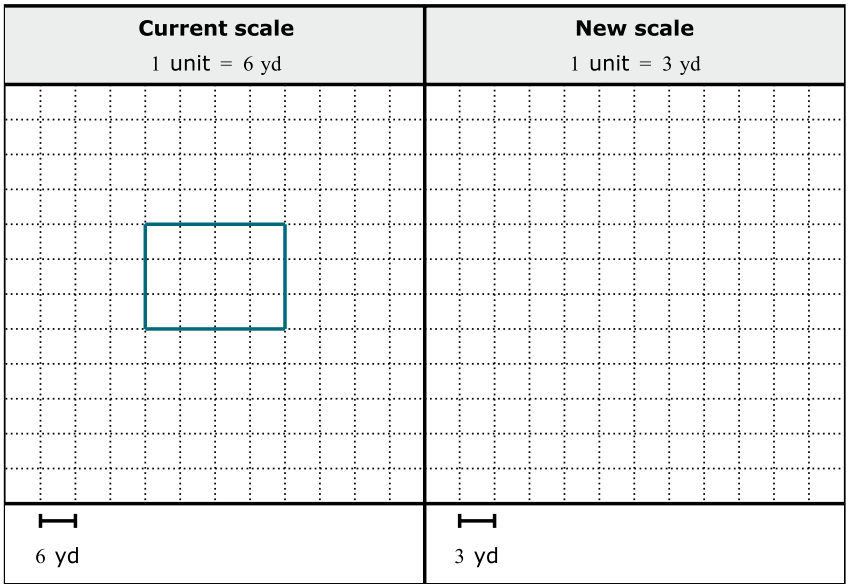
A scale drawing for a plot of land is shown below.
In the drawing, 1 cm represents 3 m.

Assuming the backyard is rectangular, find the area of the real backyard.



Question 4 of 34

The figure below was made with a scale of 1 unit $=6$ yd.
Draw the figure with a new scale of 1 unit $=3$ yd.
You can place your figure anywhere on the grid on the right.



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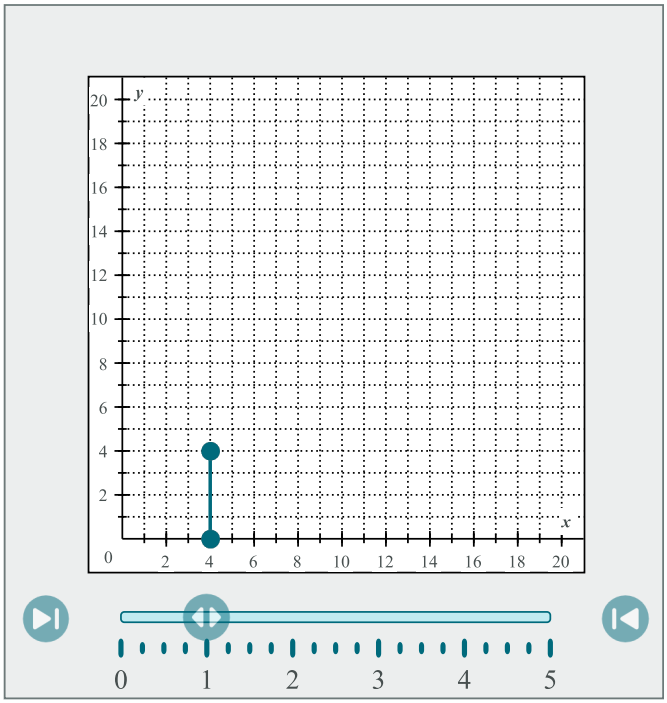
Question 5 of 34

For the figure below, do a dilation centered at the origin with a scale factor of 4.

Then, give the endpoints for both the original figure and the final figure.



Endpoints of original figure:
Top: (____, ____) Bottom: (____, ____)

Endpoints of final figure:
Top: (____, ____) Bottom: (____, ____)

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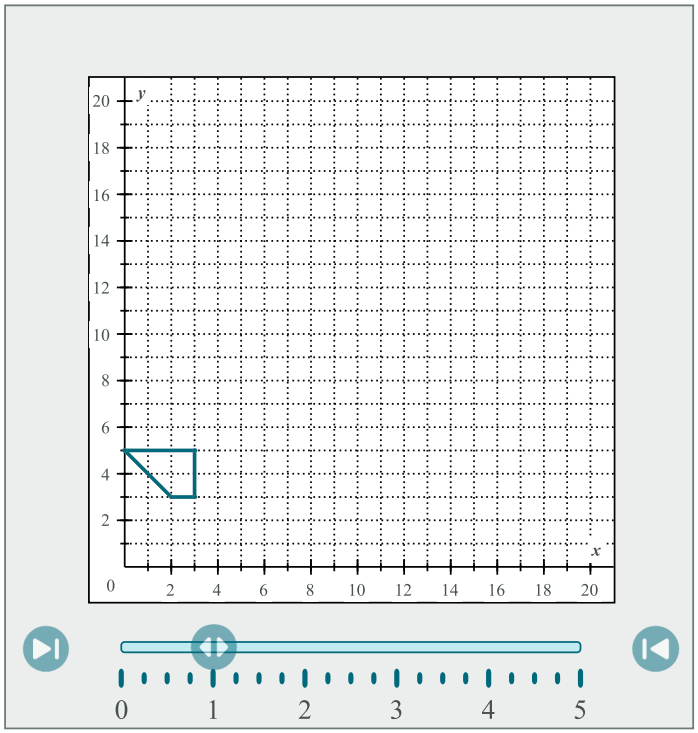
Question 6 of 34

For the figure, do a dilation centered at the origin with a scale factor of 3. Then answer each part.



(a) Fill in the blanks.

Shortest side length of the original figure: _____ units
Shortest side length of the final figure: _____ units

(b) Fill in the blank to make a true statement.

Shortest side length of the final figure $=$ _____ $\times$ Shortest side length of the original figure

(c) True or False?

A dilation with a positive scale factor *greater* than 1 gives a final figure *smaller* than the original figure.

- True
- False

(d) True or False?

The original figure and the final figure are not similar.

- True
- False

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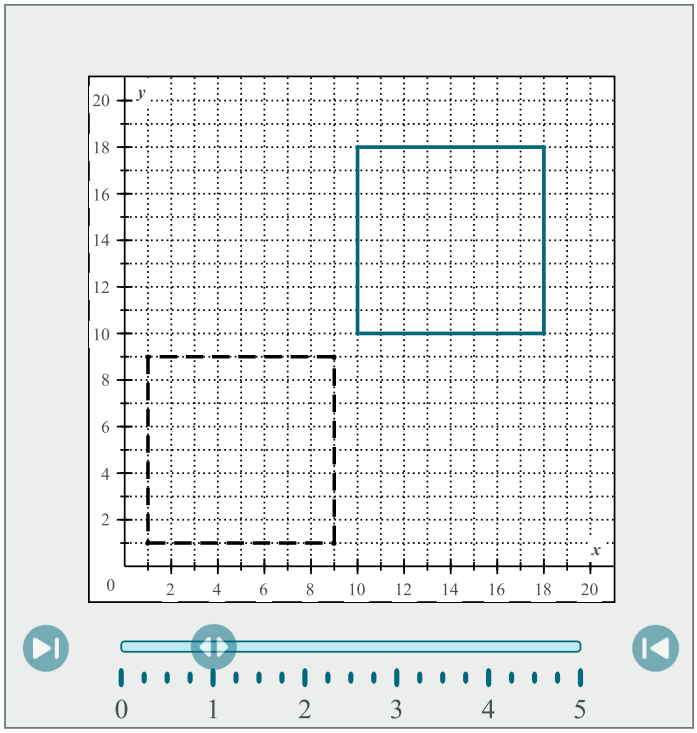
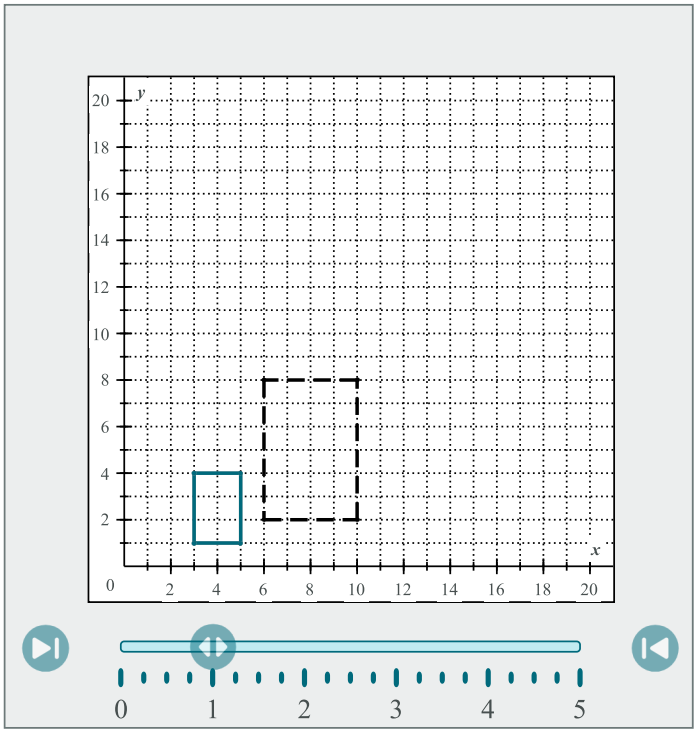
In each part, dilate the solid figure exactly onto the dashed figure.
(Note that the dilations are centered at the origin.)

(a)



(b)



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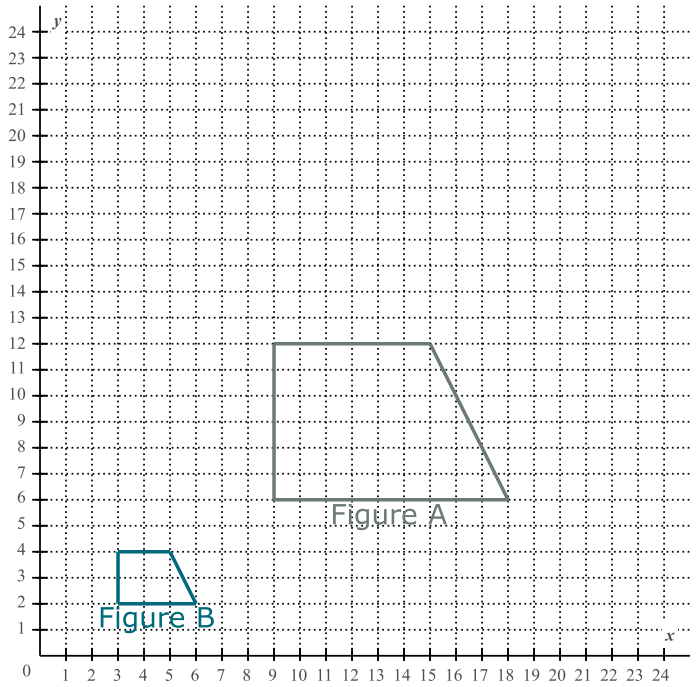
Question 8 of 34

A dilation centered at the origin is applied to Figure A. The result is Figure B.



What is the scale factor of the dilation? Write your answer as a whole number or a fraction in simplest form.

_____

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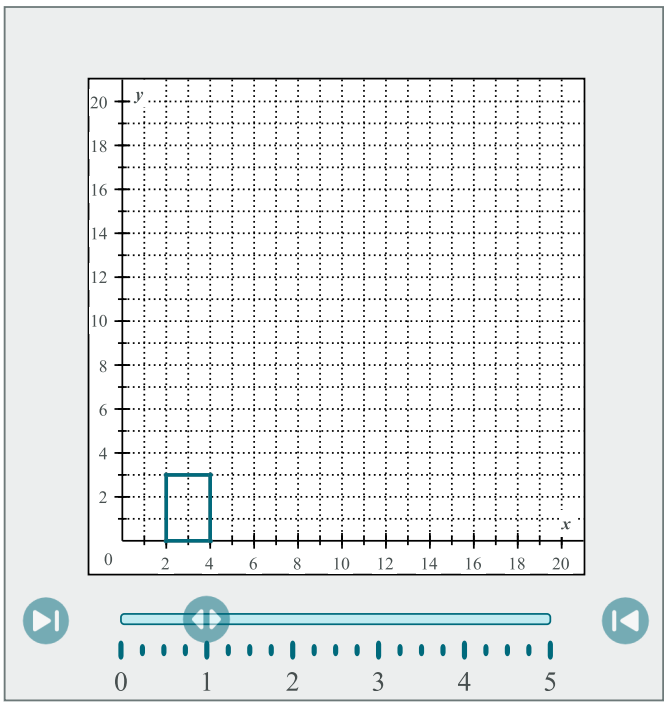
Question 9 of 34

For the figure below, do a dilation centered at the origin with a scale factor of 3.

Then answer the questions.



(a) Find the following.

Area of original figure: _____ square units
Area of final figure: _____ square units

(b) Fill in the blank to make a true statement.

Area of final figure $=$ _____ $\times$ Area of original figure

(c) True or False?

The original figure and the final figure are similar.

○ True ○ False

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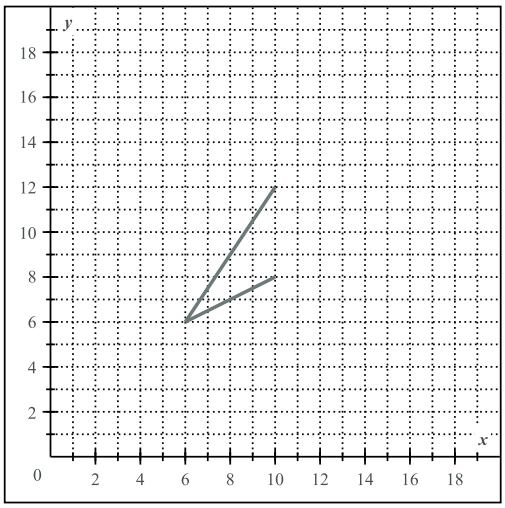
Question 10 of 34

Draw the image of the following figure after a dilation centered at the origin with a scale factor of $\frac{3}{2}$.

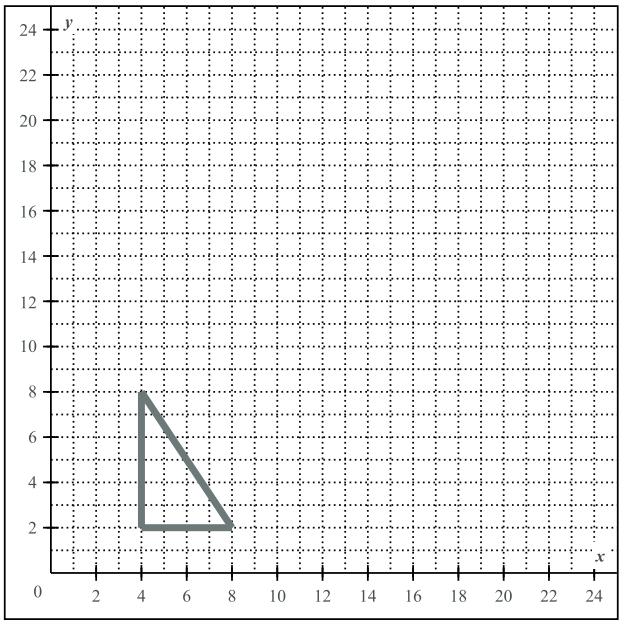


Question 11 of 34

(a) Draw each of the images described in the steps below.

Step 1: First, starting with the given figure, perform a dilation centered at the origin with a scale factor of 3.

Step 2: Then from the resulting figure, perform a dilation centered at the origin with a scale factor of $\frac{1}{2}$.



(b) Fill in the blank to describe the situation.

Performing Steps 1 and 2 is equivalent to a single dilation centered at the origin with a scale factor of _____.

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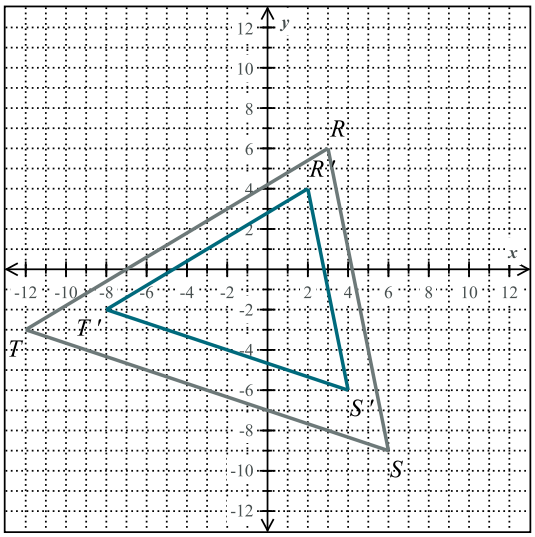
A dilation centered at the origin with a scale factor of $\frac{2}{3}$ is applied to $\triangle RST$.

The result is $\triangle R'S'T'$, as shown below.



(a)The arrows below show that the coordinates on the left are mapped to the coordinates on the right. Fill in the blanks to give the coordinates after the dilation.

original coordinates → final coordinates

$R(3, 6) \rightarrow R'$(_____, _____)

$S(6, -9) \rightarrow S'$(_____, _____)

$T(-12, -3) \rightarrow T'$(_____, _____)

(b)Choose the general rule below that describes the dilation mapping $\triangle RST$ to $\triangle R'S'T'$.

- $(x, y) \rightarrow \left(\frac{3}{2}x, \frac{2}{3}y\right)$
- $(x, y) \rightarrow \left(\frac{2}{3}x, \frac{2}{3}y\right)$
- $(x, y) \rightarrow \left(\frac{3}{2}x, \frac{3}{2}y\right)$
- $(x, y) \rightarrow \left(\frac{2}{3}y, \frac{2}{3}x\right)$
- $(x, y) \rightarrow \left(\frac{2}{3}x, y\right)$
- $(x, y) \rightarrow \left(\frac{2}{3}x, \frac{3}{2}y\right)$
- $(x, y) \rightarrow \left(x, \frac{2}{3}y\right)$
- $(x, y) \rightarrow \left(\frac{3}{2}y, \frac{3}{2}x\right)$

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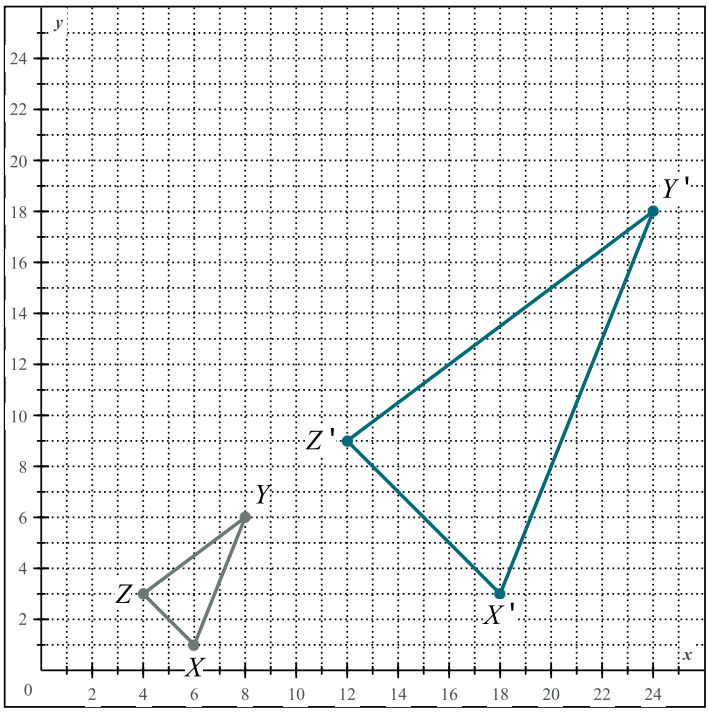
Starting with $\triangle XYZ$, a dilation centered at the origin is performed.
The result is $\triangle X'Y'Z'$, as shown.

Note that you can use the graphing tools to help you answer the questions below.

Note that $\triangle XYZ$ has vertices $X(6,1)$, $Y(8,6)$, and $Z(4,3)$.
Also, note that $\triangle X'Y'Z'$ has vertices $X'(18,3)$, $Y'(24,18)$, and $Z'(12,9)$.



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(a) Complete the following.

The line through Y and Z

- passes through the center of dilation.
- does not pass through the center of dilation.

The line through X and Y

- passes through the center of dilation.
- does not pass through the center of dilation.

(b) Find each slope below.

slope of $\overleftrightarrow{YZ}$: slope of $\overleftrightarrow{XY}$:

slope of $\overleftrightarrow{Y'Z'}$: slope of $\overleftrightarrow{X'Y'}$:

(c) The lines $\overleftrightarrow{YZ}$ and $\overleftrightarrow{Y'Z'}$ are

- parallel and distinct lines.
- the same line.
- perpendicular lines.

The lines $\overleftrightarrow{XY}$ and $\overleftrightarrow{X'Y'}$ are

- parallel and distinct lines.
- the same line.
- perpendicular lines.

(d) A dilation takes a line not passing through the center of the dilation

- a parallel and distinct line.
- the same line.
- a perpendicular line.

A dilation takes a line passing through the center of the dilation

- a parallel and distinct line.
- the same line.
- a perpendicular line.

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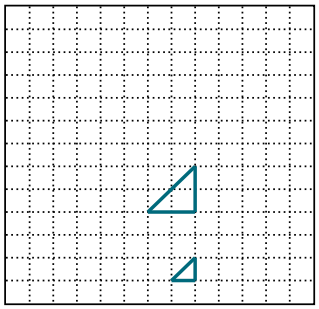
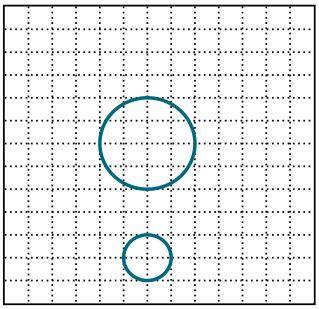
Question 14 of 34

Which pairs of figures are congruent? Which pairs are similar?



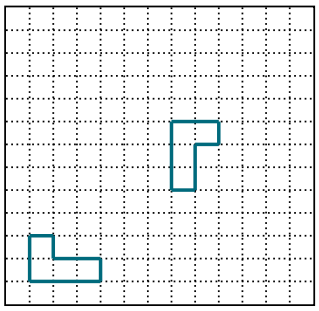
Congruent? ◯ Yes ◯ No

Similar? ◯ Yes ◯ No



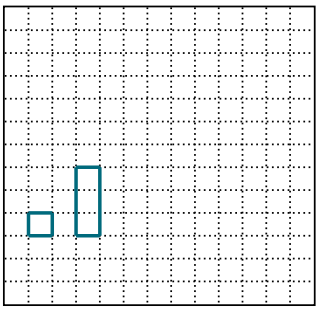
Congruent? ◯ Yes ◯ No

Similar? ◯ Yes ◯ No



Congruent? ◯ Yes ◯ No

Similar? ◯ Yes ◯ No

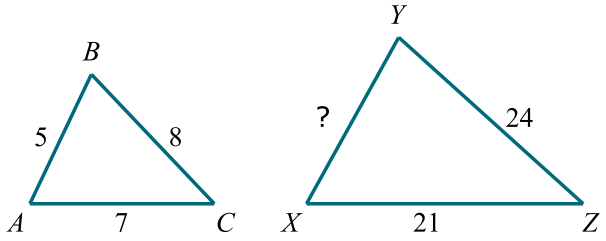


Congruent? ◯ Yes ◯ No

Similar? ◯ Yes ◯ No

Question 15 of 34

$\triangle ABC$ and $\triangle XYZ$ are similar. Find the missing side length.



(The triangles are not drawn to scale.)

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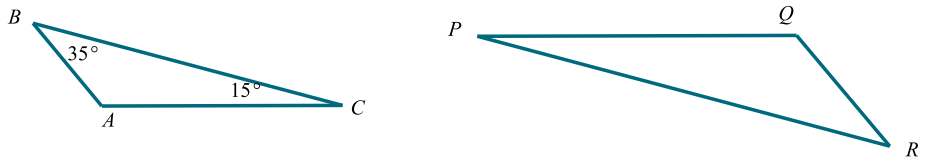
The two triangles below are similar.
Also, $m\angle C = 15°$ and $m\angle B = 35°$ as shown below.
Find $m\angle P$, $m\angle Q$, and $m\angle R$.

Assume the triangles are accurately drawn.



$m\angle P =$ ____°
$m\angle Q =$ ____°
$m\angle R =$ ____°

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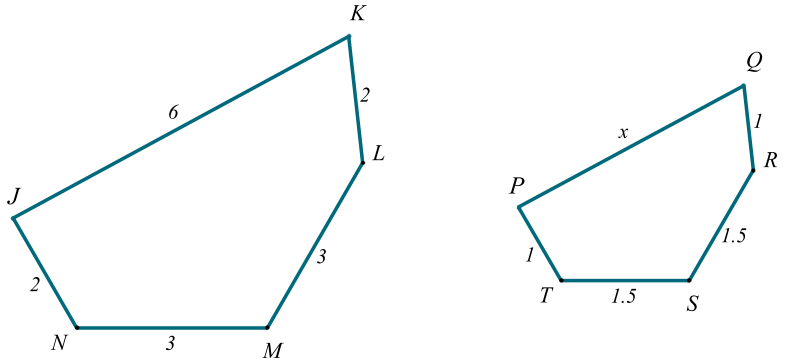
The pentagons $JKLMN$ and $PQRST$ are similar.
Find the length x of $\overline{PQ}$.



$x =$ ☐

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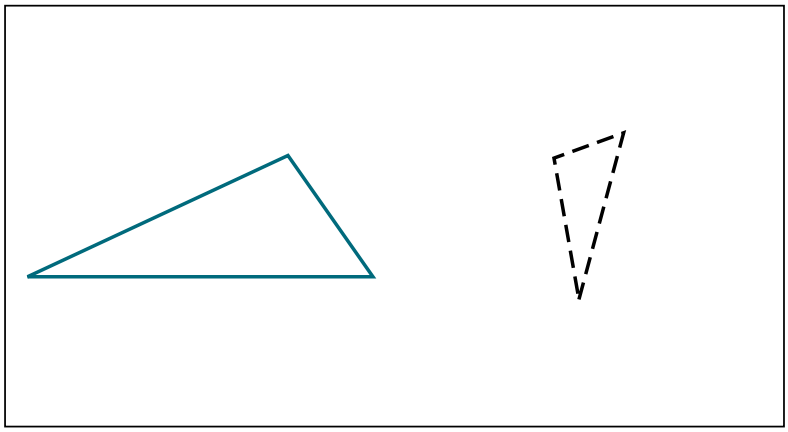
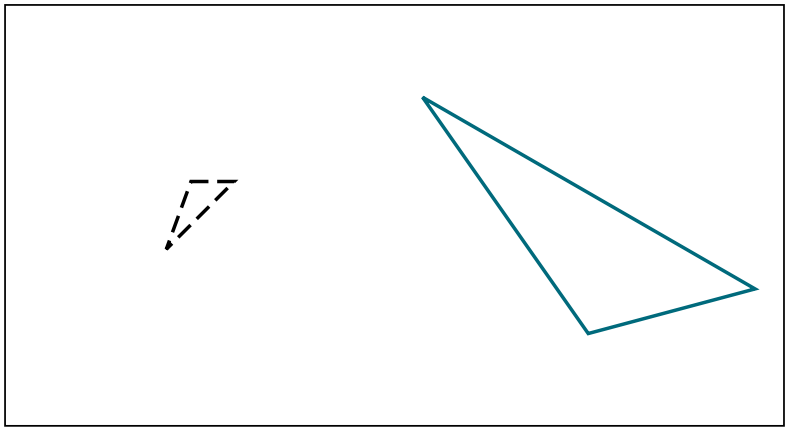
In each part, use the tools to move the solid triangle exactly onto the dashed triangle.

If this is not possible, click on "Not possible".

(a)



(b)



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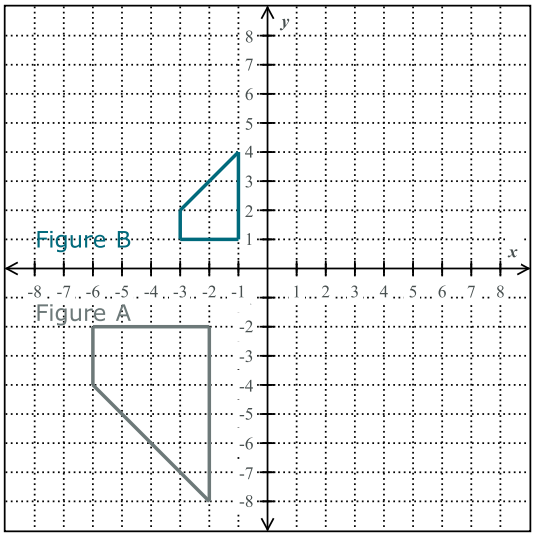
Answer the questions about Figure A and Figure B below.



(a) Which sequence(s) of transformations will map Figure A onto Figure B exactly? Choose all that apply.

- [ ] Reflect Figure A over the x-axis, and then dilate that result with scale factor $\frac{1}{2}$ centered at the origin.
- [ ] Rotate Figure A counterclockwise $90°$ about the origin, and then dilate that result with scale factor $\frac{1}{2}$ centered at the origin.
- [ ] Dilate Figure A with scale factor $\frac{1}{2}$ centered at the origin, and then reflect that result over the x-axis.
- [ ] Dilate Figure A with scale factor $\frac{1}{2}$ centered at the origin, and then translate that result to the right 5 units.
- [ ] None of these

(b) Are Figure A and Figure B similar?

○ Yes ○ No

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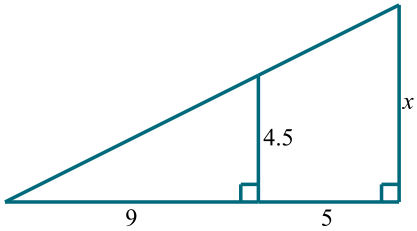
Find the length x.



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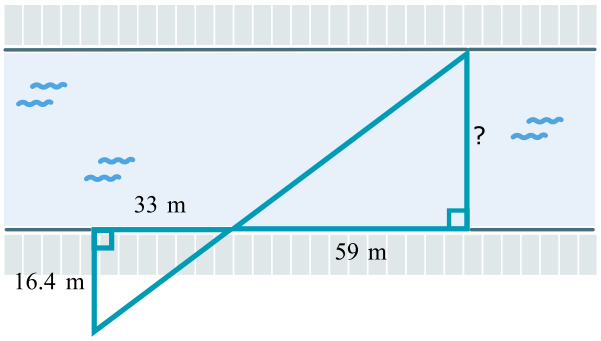
Jose wants to measure the width of a river. He marks off two right triangles, as shown in the figure. The base of the larger triangle has a length of $59\ \mathrm{m}$, and the base of the smaller triangle has a length of $33\ \mathrm{m}$. The height of the smaller triangle is $16.4\ \mathrm{m}$. How wide is the river? Round your answer to the nearest meter. (The figure is not drawn to scale.)



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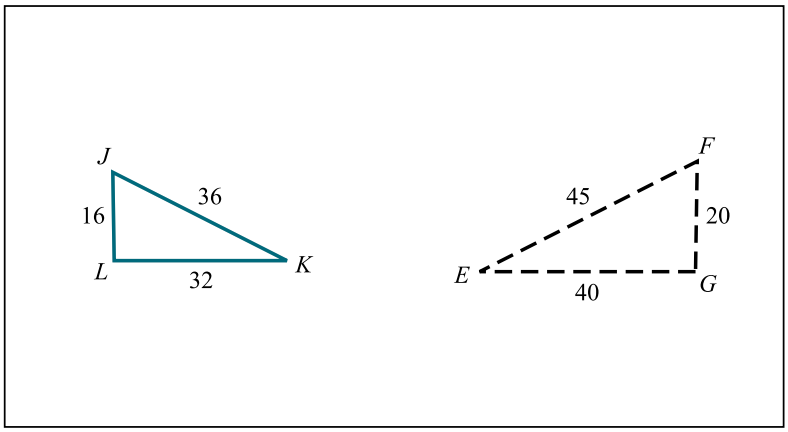
For the triangles below, use the tools to move the solid triangle exactly onto the dashed one.

Then answer the parts below.



(a) Complete the congruence statements below.

$\angle J \cong$ (Choose one)

$\angle K \cong$ (Choose one)

$\angle L \cong$ (Choose one)

(b) Find the following ratios.
Write your answers as fractions in simplest form.

$\frac{FE}{JK} = \square$ $\frac{EG}{KL} = \square$ $\frac{FG}{JL} = \square$

(c) Select the true statement.

- $\triangle JKL$ and $\triangle FEG$ are similar.
- $\triangle JKL$ and $\triangle FEG$ are *not* similar.

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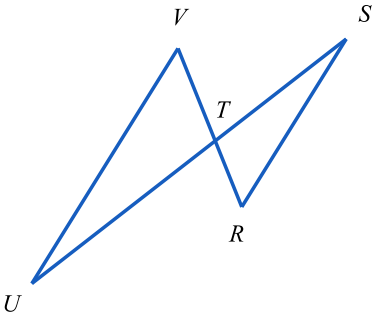
Use the given information to prove that $\triangle RST \sim \triangle VUT$.



Given: $\angle R \cong \angle V$

Prove: $\triangle RST \sim \triangle VUT$

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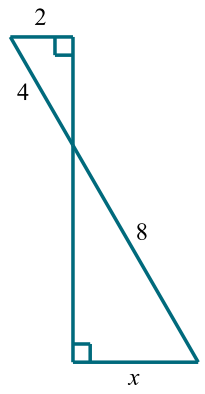
Find the length x.



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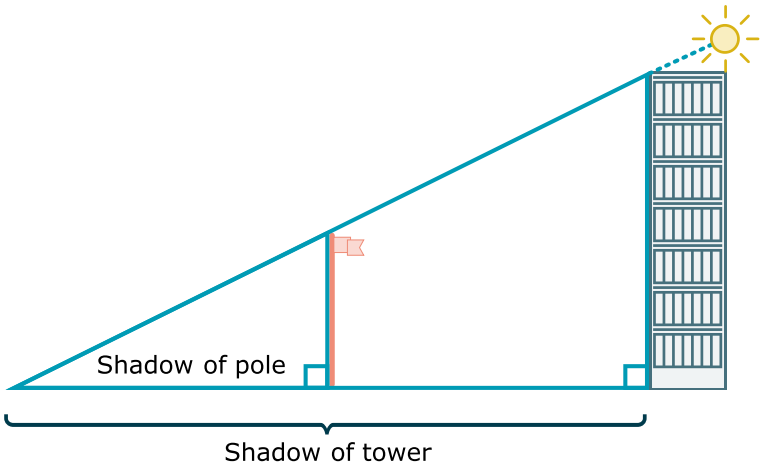
A woman wants to measure the height of a nearby tower. She places a $5\ \text{ft}$ pole in the shadow of the tower so that the shadow of the pole is exactly covered by the shadow of the tower. The total length of the tower's shadow is $189\ \text{ft}$, and the pole casts a shadow that is $3.25\ \text{ft}$ long. How tall is the tower? Round your answer to the nearest foot. (The figure is not drawn to scale.)



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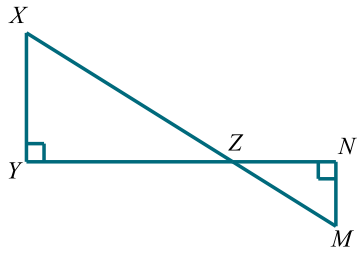
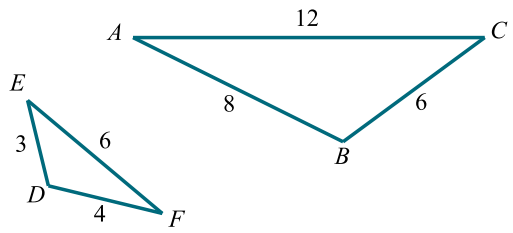
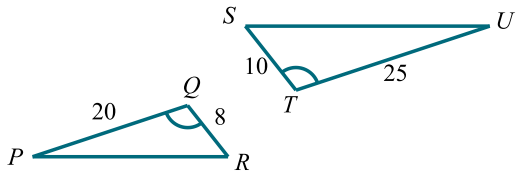
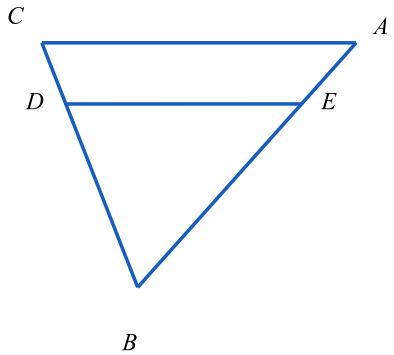
Decide whether the triangles are similar. If they are, write a similarity statement and state the reason justifying the similarity.

If necessary, you may learn what the markings on a figure indicate.

	○ Not similar or not necessarily similar ○ Similar: $\triangle XYZ \sim \triangle$____, by the ○ Angle-Angle (AA) Similarity Property ○ Side-Side-Side (SSS) Similarity Property ○ Side-Angle-Side (SAS) Similarity Property
	○ Not similar or not necessarily similar ○ Similar: $\triangle ABC \sim \triangle$____, by the ○ Angle-Angle (AA) Similarity Property ○ Side-Side-Side (SSS) Similarity Property ○ Side-Angle-Side (SAS) Similarity Property
	○ Not similar or not necessarily similar ○ Similar: $\triangle STU \sim \triangle$____, by the ○ Angle-Angle (AA) Similarity Property ○ Side-Side-Side (SSS) Similarity Property ○ Side-Angle-Side (SAS) Similarity Property

Question 27 of 34

Use the given information to prove that $\triangle ABC \sim \triangle EBD$.



Given: $\angle ACB \cong \angle EDB$

Prove: $\triangle ABC \sim \triangle EBD$

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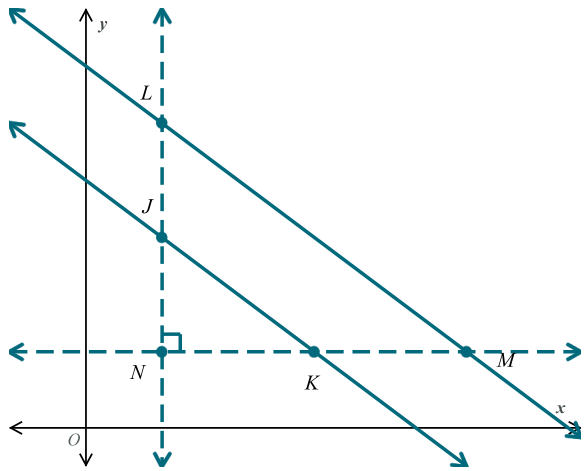
Question 28 of 34

In the figure below, $\overleftrightarrow{JK}$ and $\overleftrightarrow{LM}$ are parallel.



Complete the following.

(a) Find the slope of $\overleftrightarrow{JK}$ and the slope of $\overleftrightarrow{LM}$.

slope of $\overleftrightarrow{JK}$: $-\frac{NK}{NJ}$ $\frac{NK}{NJ}$ $-\frac{NJ}{NK}$ $\frac{NJ}{NK}$

slope of $\overleftrightarrow{LM}$: $\frac{NM}{NL}$ $-\frac{NL}{NM}$ $-\frac{NM}{NL}$ $\frac{NL}{NM}$

(b) It can be shown that $\triangle NJK \sim \triangle NLM$.

Based on this, choose the ratio that is equal to $\frac{NJ}{NK}$.

$\frac{NM}{NL}$ $\frac{LM}{NM}$ $\frac{NL}{LM}$ $\frac{NL}{NM}$

(c) Using the results above, choose the correct statement below.

- slope of $\overleftrightarrow{JK}$ = slope of $\overleftrightarrow{LM}$
- slope of $\overleftrightarrow{JK}$ · slope of $\overleftrightarrow{LM}$ $=1$
- slope of $\overleftrightarrow{JK}$ · slope of $\overleftrightarrow{LM}$ $=-1$
- slope of $\overleftrightarrow{JK}$ $=-$slope of $\overleftrightarrow{LM}$

(d) The result in part (c) is an example of the following rule for any two non-vertical parallel lines.

- The slopes of the two lines are reciprocals.
- The slopes of the two lines are the same.
- The slopes of the two lines are negative reciprocals.
- The slopes of the two lines are opposites.

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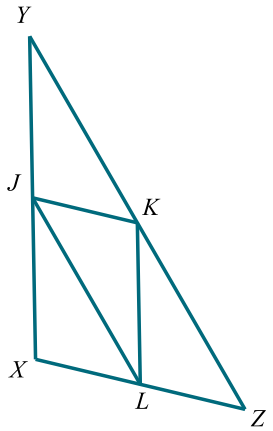
Question 29 of 34

In the figure below, points J, K, and L are the midpoints of the sides of $\triangle XYZ$.

Suppose $YZ = 78$, $JK = 16$, and $XY = 70$.
Find the following lengths.



$XZ =$ _____

$LZ =$ _____

$JL =$ _____

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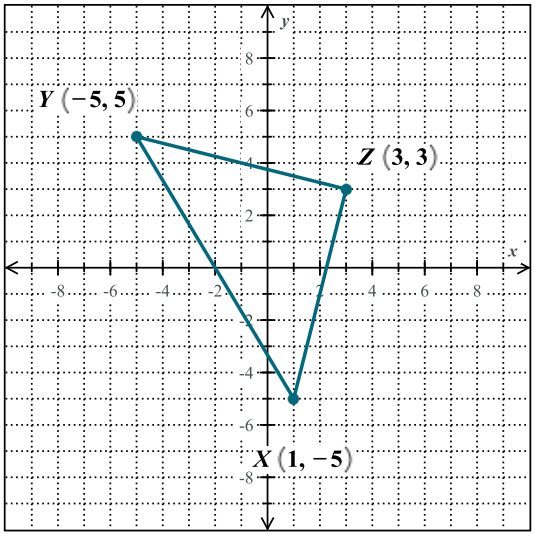
Question 30 of 34

Consider $\triangle XYZ$ below.



Note that $\triangle XYZ$ has vertices $X(1,\ -5)$, $Y(-5,\ 5)$, and $Z(3,\ 3)$.

Suppose that R is the midpoint of $\overline{XZ}$ and that S is the midpoint of $\overline{YZ}$.

Answer the following.

(a)Find the coordinates of R and S.

R (__, __) S (__, __)

(b)Find the following slopes. Write each answer as an integer or fraction in simplest form.

Slope of $\overline{RS}$ = __ Slope of $\overline{XY}$ = __

(c)Find the following lengths. Give exact answers in simplest form.

Length of $\overline{RS}$ = __ Length of $\overline{XY}$ = __

(d)The relationship between $\overline{RS}$ and $\overline{XY}$ is an example of the following rules.

A segment that joins the midpoints of two sides of a triangle is

- ○ perpendicular to the third side.
- ○ parallel to the third side.
- ○ neither parallel nor perpendicular to the third side.

The length of a segment that joins the midpoints of two sides of a triangle is

- ○ the same length as the third side.
- ○ twice the length of the third side.
- ○ half the length of the third side.
- ○ not related to the length of the third side.

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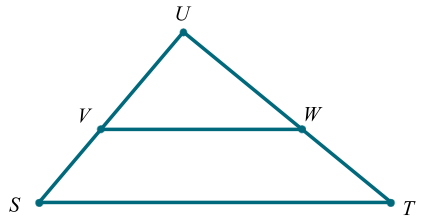
In $\triangle STU$, $\overline{ST} \parallel \overline{VW}$. Given that $US = 35$, $UV = 20$, and $UT = 42$, find UW.



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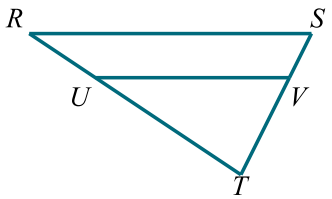
Use the given information to complete the proof of the following theorem.

If a line parallel to one side of a triangle intersects the other two sides, then it divides the two sides proportionally.



Given: $\overline{SR} \parallel \overline{VU}$

Prove: $\dfrac{SV}{VT} = \dfrac{RU}{UT}$

	Statement	Reason
1	$\overline{SR} \parallel \overline{VU}$	Given
2	$\angle TSR \cong \angle$	If lines $\parallel$, then corr. $\angle$s $\cong$
3	$\angle T \cong \angle T$	_____
4	$\triangle RST \sim \triangle$	AA Similarity Property
5	$\dfrac{ST}{VT} = \dfrac{RT}{UT}$	_____
6	$ST = SV +$	Segment Addition Property
7	$RT = RU +$	Segment Addition Property
8	$\dfrac{SV + VT}{VT} = \dfrac{RU + UT}{UT}$	_____
9	$\dfrac{SV}{VT} + \dfrac{VT}{VT} = \dfrac{RU}{UT} + \dfrac{UT}{UT}$	Fraction Algebra
10	$\dfrac{SV}{VT} + 1 = \dfrac{RU}{UT} + 1$	Fraction Algebra
11	$\dfrac{SV}{VT} = \dfrac{\quad}{UT}$	_____

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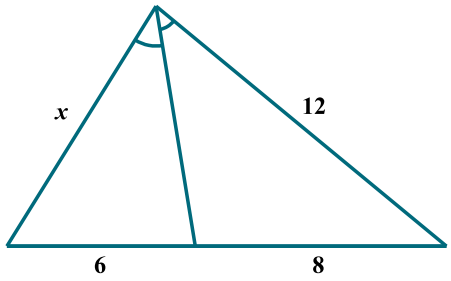
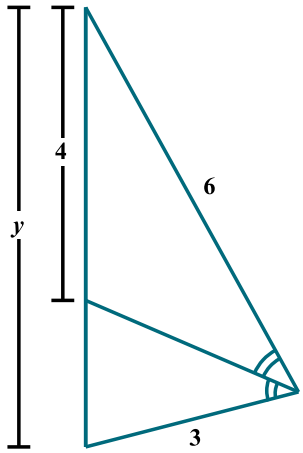
Find the value of each variable.
Simplify your answers as much as possible.





x = _____

y = _____

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Simplify.

$$\frac{36xy}{6wxy}$$

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Class Name: **ALEKS Integrated Mathematics II** Number of Questions: **34**

Question 1 of 34

(a) The length of the real house is 27 feet. What is the length of the house in the scale drawing? 3 inches
(b) In the scale drawing, the height of the house is 8 inches. What is the height of the real house? 72 feet

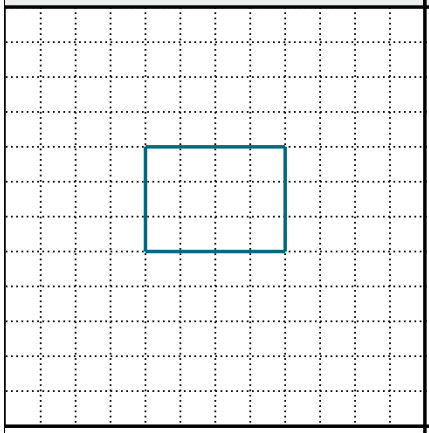
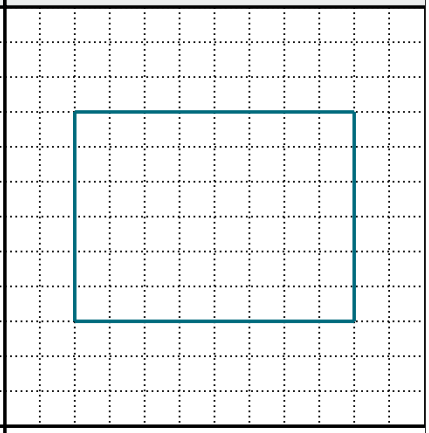
Question 2 of 34

Scale factor: $\frac{1}{5}$

Question 3 of 34

Area of the real backyard: $108\ \mathrm{m}^2$

Question 4 of 34

Current scale 1 unit $=$ 6 yd	New scale 1 unit $=$ 3 yd
	
6 yd	3 yd

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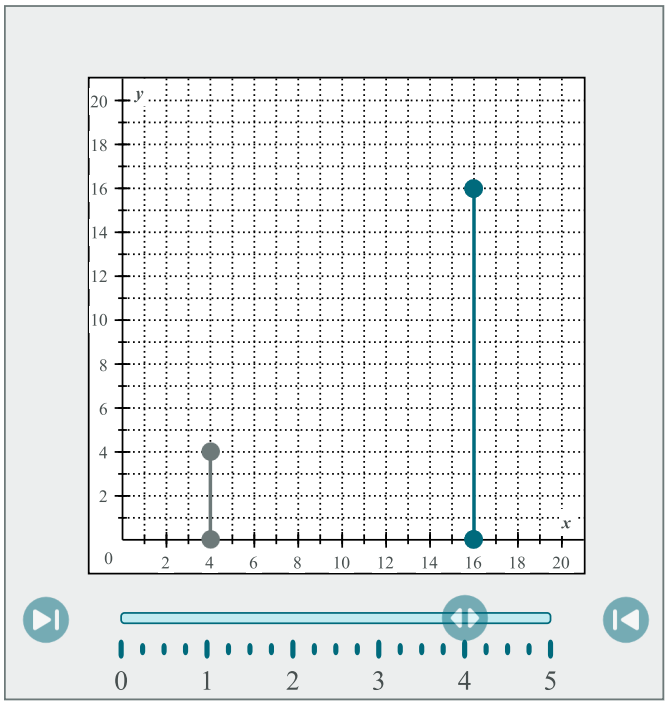
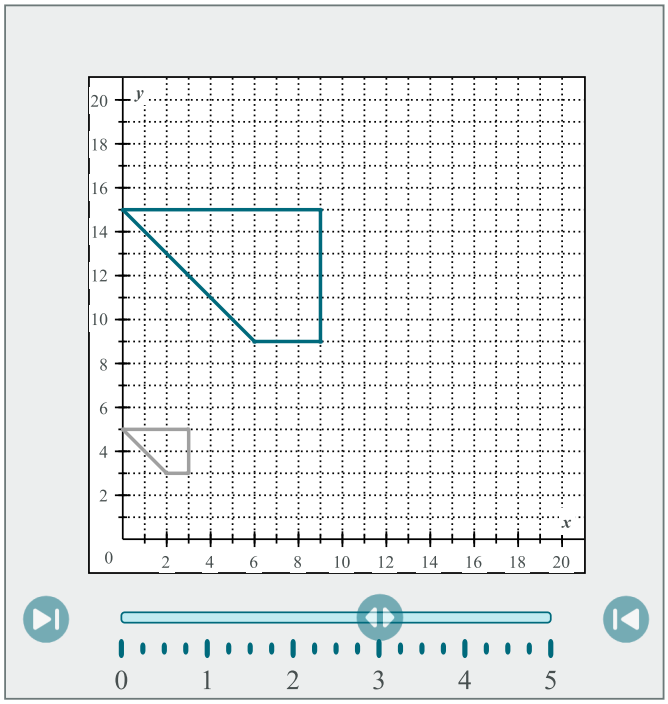


Endpoints of original figure:
Top: $(4, 4)$ Bottom: $(4, 0)$

Endpoints of final figure:
Top: $(16, 16)$ Bottom: $(16, 0)$

Question 6 of 34



(a) Fill in the blanks.

Shortest side length of the original figure: 1 units
Shortest side length of the final figure: 3 units

(b) Fill in the blank to make a true statement.

Shortest side length of the final figure $= 3 \times$ Shortest side length of the original figure

(c) True or False?

A dilation with a positive scale factor *greater* than 1 gives a final figure *smaller* than the original figure.

○ True ◉ False

(d) True or False?

The original figure and the final figure are not similar.

○ True ◉ False

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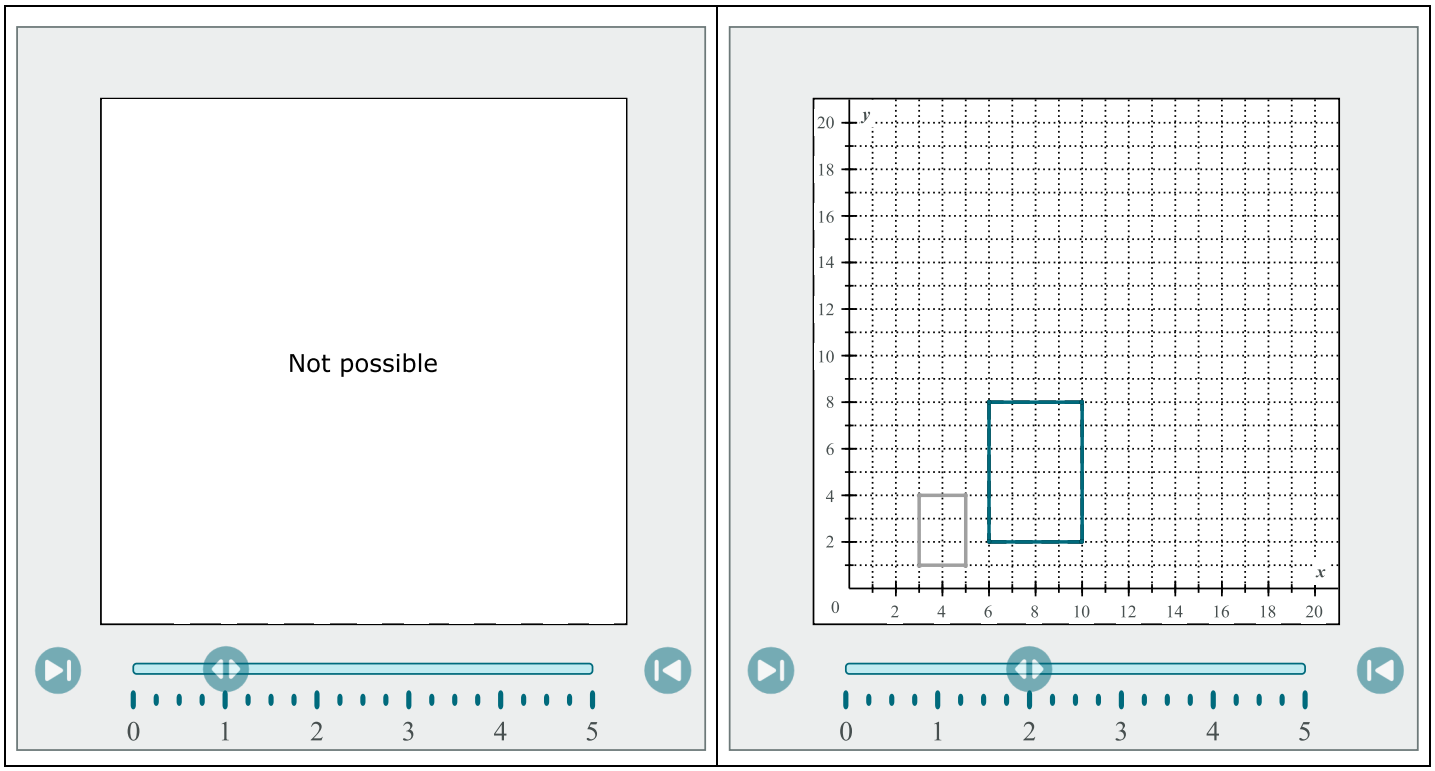
Question 7 of 34



Question 8 of 34

$\frac{1}{3}$

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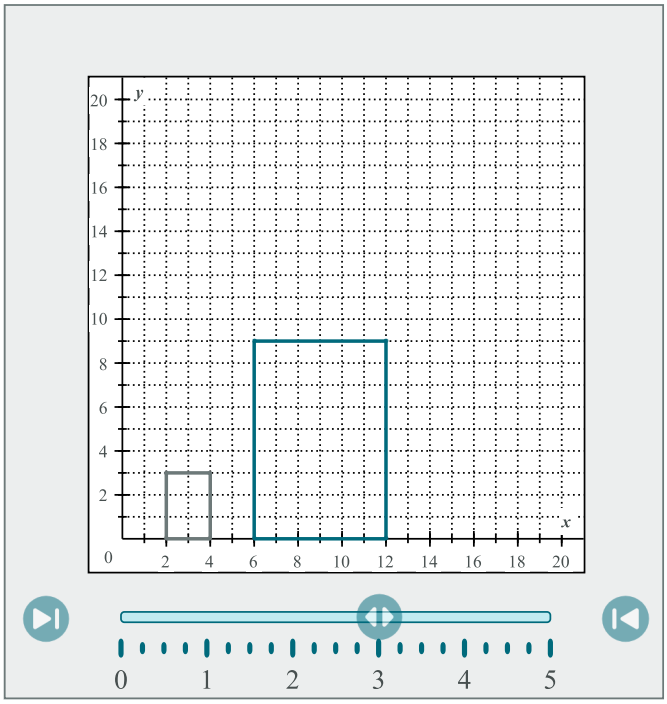

(a) Find the following.

Area of original figure: 6 square units

Area of final figure: 54 square units

(b) Fill in the blank to make a true statement.

Area of final figure $=9\times$ Area of original figure

(c) True or False?

The original figure and the final figure are similar.

- (●) True
- ( ) False

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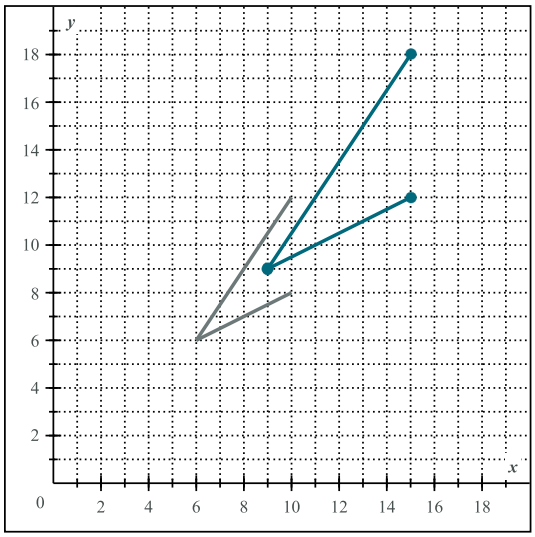
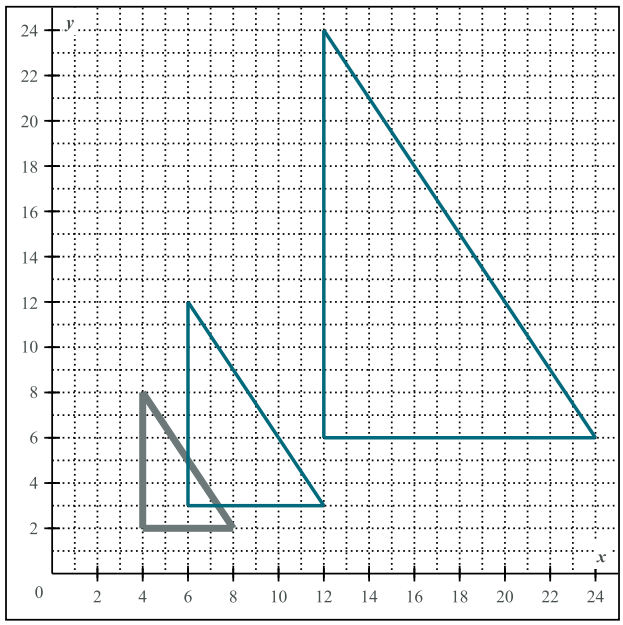
Question 10 of 34



Question 11 of 34

(a)



(b) Performing Steps 1 and 2 is equivalent to a single dilation centered at the origin with a scale factor of $\frac{3}{2}$.

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(a) The arrows below show that the coordinates on the left are mapped to the coordinates on the right. Fill in the blanks to give the coordinates after the dilation.

original coordinates $\longrightarrow$ final coordinates

$$R(3,\ 6) \rightarrow R'(2,\ 4)$$

$$S(6,\ -9) \rightarrow S'(4,\ -6)$$

$$T(-12,\ -3) \rightarrow T'(-8,\ -2)$$

(b) Choose the general rule below that describes the dilation mapping $\triangle RST$ to $\triangle R'S'T'$.

- ○ $(x, y) \rightarrow \left(\frac{3}{2}x,\ \frac{2}{3}y\right)$
- ◉ $(x, y) \rightarrow \left(\frac{2}{3}x,\ \frac{2}{3}y\right)$
- ○ $(x, y) \rightarrow \left(\frac{3}{2}x,\ \frac{3}{2}y\right)$
- ○ $(x, y) \rightarrow \left(\frac{2}{3}y,\ \frac{2}{3}x\right)$
- ○ $(x, y) \rightarrow \left(\frac{2}{3}x,\ y\right)$
- ○ $(x, y) \rightarrow \left(\frac{2}{3}x,\ \frac{3}{2}y\right)$
- ○ $(x, y) \rightarrow \left(x,\ \frac{2}{3}y\right)$
- ○ $(x, y) \rightarrow \left(\frac{3}{2}y,\ \frac{3}{2}x\right)$

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Question 13 of 34

(a) Complete the following.

The line through Y and Z

- passes through the center of dilation.

The line through X and Y

- does not pass through the center of dilation.

(b) Find each slope below.

slope of $\overleftrightarrow{YZ}$: $\frac{3}{4}$ slope of $\overleftrightarrow{XY}$: $\frac{5}{2}$

slope of $\overleftrightarrow{Y'Z'}$: $\frac{3}{4}$ slope of $\overleftrightarrow{X'Y'}$: $\frac{5}{2}$

(c) The lines $\overleftrightarrow{YZ}$ and $\overleftrightarrow{Y'Z'}$ are

- parallel and distinct lines.

The lines $\overleftrightarrow{XY}$ and $\overleftrightarrow{X'Y'}$ are

- the same line.

(d) A dilation takes a line not passing through the center of the dilation

- the same line.

A dilation takes a line passing through the center of the dilation

- the same line.

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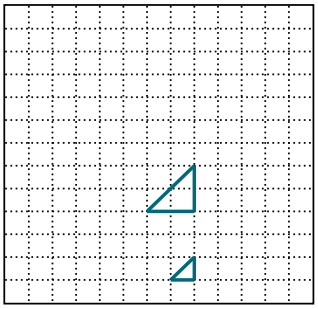
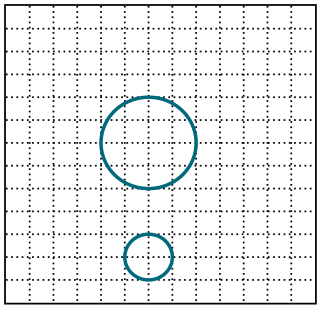
Question 14 of 34



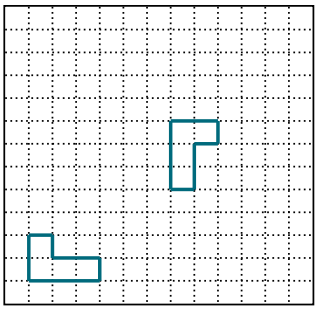
Congruent? ○ Yes ◉ No

Similar? ◉ Yes ○ No



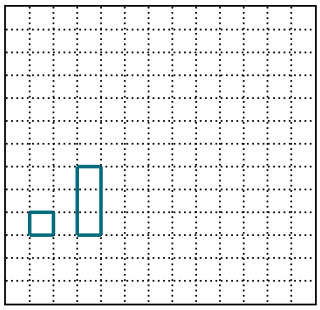
Congruent? ○ Yes ◉ No

Similar? ◉ Yes ○ No



Congruent? ◉ Yes ○ No

Similar? ◉ Yes ○ No



Congruent? ○ Yes ◉ No

Similar? ○ Yes ◉ No

Question 15 of 34

15

Question 16 of 34

$m\angle P = 15°$

$m\angle Q = 130°$

$m\angle R = 35°$

Question 17 of 34

$x = 3$

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Question 18 of 34

(a)

(b)

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Question 19 of 34

(a) Which sequence(s) of transformations will map Figure A onto Figure B exactly? Choose all that apply.

- [x] Reflect Figure A over the x-axis, and then dilate that result with scale factor $\frac{1}{2}$ centered at the origin.
- [ ] Rotate Figure A counterclockwise $90°$ about the origin, and then dilate that result with scale factor $\frac{1}{2}$ centered at the origin.
- [x] Dilate Figure A with scale factor $\frac{1}{2}$ centered at the origin, and then reflect that result over the x-axis.
- [ ] Dilate Figure A with scale factor $\frac{1}{2}$ centered at the origin, and then translate that result to the right 5 units.
- [ ] None of these

(b) Are Figure A and Figure B similar?

- (●) Yes
- ( ) No

Question 20 of 34

7

Question 21 of 34

29 m

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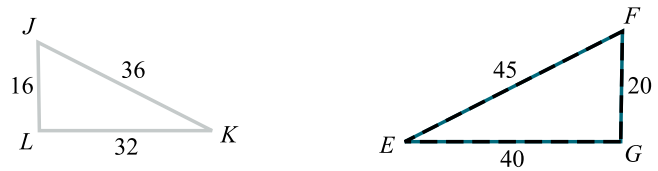

(a) Complete the congruence statements below.

$\angle J \cong$ $\angle F$

$\angle K \cong$ $\angle E$

$\angle L \cong$ $\angle G$

(b) Find the following ratios.
Write your answers as fractions in simplest form.

$\frac{FE}{JK} = \frac{5}{4}$ $\frac{EG}{KL} = \frac{5}{4}$ $\frac{FG}{JL} = \frac{5}{4}$

(c) Select the true statement.

- (selected) $\triangle JKL$ and $\triangle FEG$ are similar.
- $\triangle JKL$ and $\triangle FEG$ are *not* similar.

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Step	Statement	Reason
1	$\angle R \cong \angle V$	Given
2	$\angle RTS \cong \angle VTU$	Vertical Angles Property
3	$\triangle RST \sim \triangle VUT$	AA Similarity Property

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4

Question 25 of 34

291 ft

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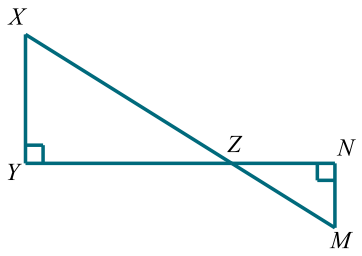
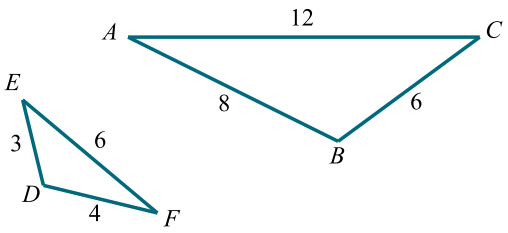
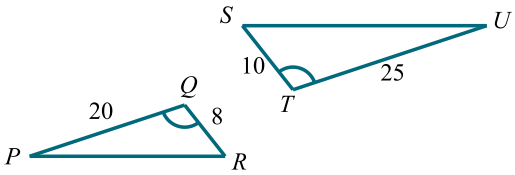
	○ Not similar or not necessarily similar ◉ Similar: $\triangle XYZ \sim \triangle MNZ$, by the ◉ Angle-Angle (AA) Similarity Property ○ Side-Side-Side (SSS) Similarity Property ○ Side-Angle-Side (SAS) Similarity Property
	○ Not similar or not necessarily similar ◉ Similar: $\triangle ABC \sim \triangle FDE$, by the ○ Angle-Angle (AA) Similarity Property ◉ Side-Side-Side (SSS) Similarity Property ○ Side-Angle-Side (SAS) Similarity Property
	○ Not similar or not necessarily similar ◉ Similar: $\triangle STU \sim \triangle RQP$, by the ○ Angle-Angle (AA) Similarity Property ○ Side-Side-Side (SSS) Similarity Property ◉ Side-Angle-Side (SAS) Similarity Property

Question 27 of 34

Step	Statement	Reason
1	$\angle ACB \cong \angle EDB$	Given
2	$\angle B \cong \angle B$	Reflexive Property
3	$\triangle ABC \sim \triangle EBD$	AA Similarity Property

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Question 28 of 34

(a) Find the slope of $\overleftrightarrow{JK}$ and the slope of $\overleftrightarrow{LM}$.

slope of $\overleftrightarrow{JK}$: ○ $-\frac{NK}{NJ}$ ○ $\frac{NK}{NJ}$ ◉ $-\frac{NJ}{NK}$ ○ $\frac{NJ}{NK}$

slope of $\overleftrightarrow{LM}$: ○ $\frac{NM}{NL}$ ◉ $-\frac{NL}{NM}$ ○ $-\frac{NM}{NL}$ ○ $\frac{NL}{NM}$

(b) It can be shown that $\triangle NJK \sim \triangle NLM$.

Based on this, choose the ratio that is equal to $\frac{NJ}{NK}$.

○ $\frac{NM}{NL}$ ○ $\frac{LM}{NM}$ ○ $\frac{NL}{LM}$ ◉ $\frac{NL}{NM}$

(c) Using the results above, choose the correct statement below.

◉ slope of $\overleftrightarrow{JK}$ = slope of $\overleftrightarrow{LM}$

○ slope of $\overleftrightarrow{JK}$ · slope of $\overleftrightarrow{LM}$ $=1$

○ slope of $\overleftrightarrow{JK}$ · slope of $\overleftrightarrow{LM}$ $=-1$

○ slope of $\overleftrightarrow{JK}$ $=-$slope of $\overleftrightarrow{LM}$

(d) The result in part (c) is an example of the following rule for any two non-vertical parallel lines.

○ The slopes of the two lines are reciprocals.

◉ The slopes of the two lines are the same.

○ The slopes of the two lines are negative reciprocals.

○ The slopes of the two lines are opposites.

Question 29 of 34

$XZ = 32$

$LZ = 16$

$JL = 39$

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Question 30 of 34

(a)Find the coordinates of R and S.

$R(2,\ -1)$ $S(-1,\ 4)$

(b)Find the following slopes. Write each answer as an integer or fraction in simplest form.

Slope of $\overline{RS} = -\frac{5}{3}$ Slope of $\overline{XY} = -\frac{5}{3}$

(c)Find the following lengths. Give exact answers in simplest form.

Length of $\overline{RS} = \sqrt{34}$ Length of $\overline{XY} = 2\sqrt{34}$

(d)The relationship between $\overline{RS}$ and $\overline{XY}$ is an example of the following rules.

A segment that joins the midpoints of two sides of a triangle is parallel to the third side.

The length of a segment that joins the midpoints of two sides of a triangle is half the length of the third side.

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$UW = 24$

Question 32 of 34

	Statement	Reason
1	$\overline{SR} \parallel \overline{VU}$	Given
2	$\angle TSR \cong \angle TVU$	If lines $\parallel$, then corr. $\angle$s $\cong$
3	$\angle T \cong \angle T$	Reflexive Property
4	$\triangle RST \sim \triangle UVT$	AA Similarity Property
5	$\frac{ST}{VT} = \frac{RT}{UT}$	Corr. sides of similar triangles are proportional
6	$ST = SV + VT$	Segment Addition Property
7	$RT = RU + UT$	Segment Addition Property
8	$\frac{SV + VT}{VT} = \frac{RU + UT}{UT}$	Substitution Property
9	$\frac{SV}{VT} + \frac{VT}{VT} = \frac{RU}{UT} + \frac{UT}{UT}$	Fraction Algebra
10	$\frac{SV}{VT} + 1 = \frac{RU}{UT} + 1$	Fraction Algebra
11	$\frac{SV}{VT} = \frac{RU}{UT}$	Addition and Subtraction Properties

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$x = 9$

$y = 6$

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$$\frac{6}{w}$$

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